

Portfolio Performance & Risk Analyzer

Historical performance and risk analysis of an equal-weight portfolio of five NSE large-cap stocks

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Data period: 7 Oct 2024 to 5 Oct 2026 (500 trading days) | Report generated: 6 Oct 2026

-21.6%

Cumulative return

-11.6%

Annualised return
(CAGR)

15.0%

Annualised volatility

-1.08

Sharpe Ratio (rf 5.26%)

-26.5%

Maximum drawdown

Headline figures for the equal-weight portfolio.

1. Objective

Measure how an equally weighted portfolio of five Indian large-cap stocks performed over the most recent two years, how much risk it carried, and where that risk came from. The analysis is descriptive: it explains what happened and why, and makes no forecast.

2. Business problem

A portfolio analytics team has to answer three questions for every portfolio it reports on. Did it make money, and how did that compare with the market? Was the return worth the risk taken? Is the portfolio as diversified as its holdings suggest? Raw return alone cannot answer the last two, so this project combines return, volatility, drawdown, correlation and risk attribution into a single, repeatable workflow.

3. Dataset and portfolio composition

Company	Yahoo ticker	Sector	Weight
Reliance Industries	RELIANCE.NS	Energy & Conglomerate	20%
TCS	TCS.NS	Information Technology	20%
HDFC Bank	HDFCBANK.NS	Banking	20%
Infosys	INFY.NS	Information Technology	20%
ICICI Bank	ICICIBANK.NS	Banking	20%
NIFTY 50 (benchmark)	^NSEI	Broad market index	n/a

Source. Daily adjusted closing prices from Yahoo Finance, downloaded with the open-source *yfinance* package (version 1.7.0) on 2026-10-06. Adjusted prices include dividends and stock splits, so returns reflect what a shareholder actually earned.

Cleaning. 500 raw dates were downloaded. 0 duplicate dates and 0 dates with no prices were removed, 6 isolated missing prices were filled with the previous day's price (at most 2 days), and 0 dates with remaining gaps were dropped. The clean dataset has 500 price dates (499 daily returns, since the first date has no previous price) from 7 Oct 2024 to 5 Oct 2026, the same dates for every security.

4. Methodology

Metric	Formula	What it tells us
Daily return	$r_t = P_t / P_{t-1} - 1$	Percentage change in price from one trading day to the next
Portfolio return	$r_p = \text{sum of } w_i \times r_i$	Weighted average of stock returns, with weights reset to 20% daily
Cumulative return	product of $(1 + r_t) - 1$	Total gain or loss over the whole period
CAGR	$(1 + \text{cumulative})^{252/N} - 1$	Constant yearly return giving the same end value
Volatility	$\text{std}(\text{daily } r) \times \text{sqrt}(252)$	Typical size of swings around the average, per year

Metric	Formula	What it tells us
Sharpe Ratio	$\text{mean}(r - r_f) / \text{std}(r) \times \text{sqrt}(252)$	Excess return earned per unit of risk
Maximum drawdown	$\min(\text{value} / \text{running peak} - 1)$	Largest fall from a previous high
Beta	$\text{cov}(r, r_m) / \text{var}(r_m)$	Sensitivity to moves in the NIFTY 50
Risk contribution	$w_i \times (\text{Cov } w)_i / \text{sigma}_p$	Part of portfolio volatility caused by each stock

Assumptions. (1) 252 trading days per year, the market convention for annualising (this dataset has about 250 NSE trading days a year). (2) Risk-free rate of 5.26% a year (91-day T-Bill cut-off yield, RBI auction of 2 Sep 2026), or 0.0203% a day. (3) Weights are held at 20% each day (daily rebalancing); buy-and-hold is shown for comparison. (4) No transaction costs or taxes. (5) Volatility uses the sample standard deviation.

5. Performance analysis

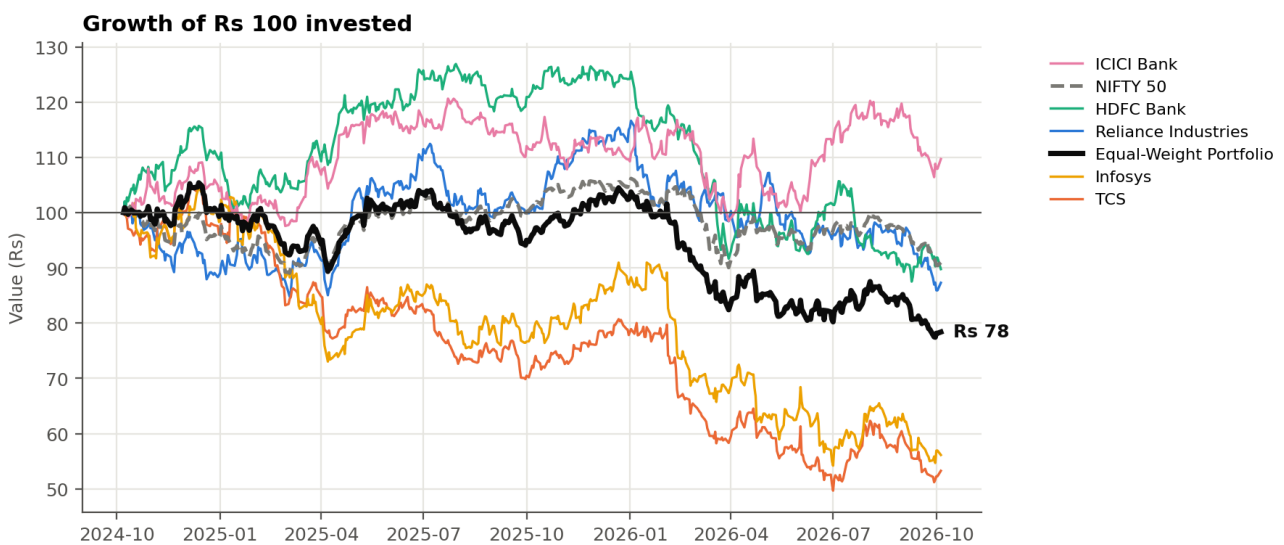


Figure 1. Value of Rs 100 invested on the first day of the period.

Security	Cumulative return	CAGR	Volatility	Sharpe	Max drawdown	Beta
Reliance Industries	-12.7%	-6.6%	20.6%	-0.48	-26.3%	1.04
TCS	-46.7%	-27.2%	24.8%	-1.36	-52.7%	0.84
HDFC Bank	-10.2%	-5.3%	19.7%	-0.44	-31.0%	1.07
Infosys	-43.9%	-25.3%	28.0%	-1.09	-48.2%	0.90
ICICI Bank	9.7%	4.8%	18.4%	0.07	-18.4%	0.90
Equal-Weight Portfolio	-21.6%	-11.6%	15.0%	-1.08	-26.5%	0.95
NIFTY 50	-9.0%	-4.7%	13.0%	-0.70	-15.2%	1.00

Table 1. Performance and risk metrics, computed from daily adjusted prices.

6. Risk analysis

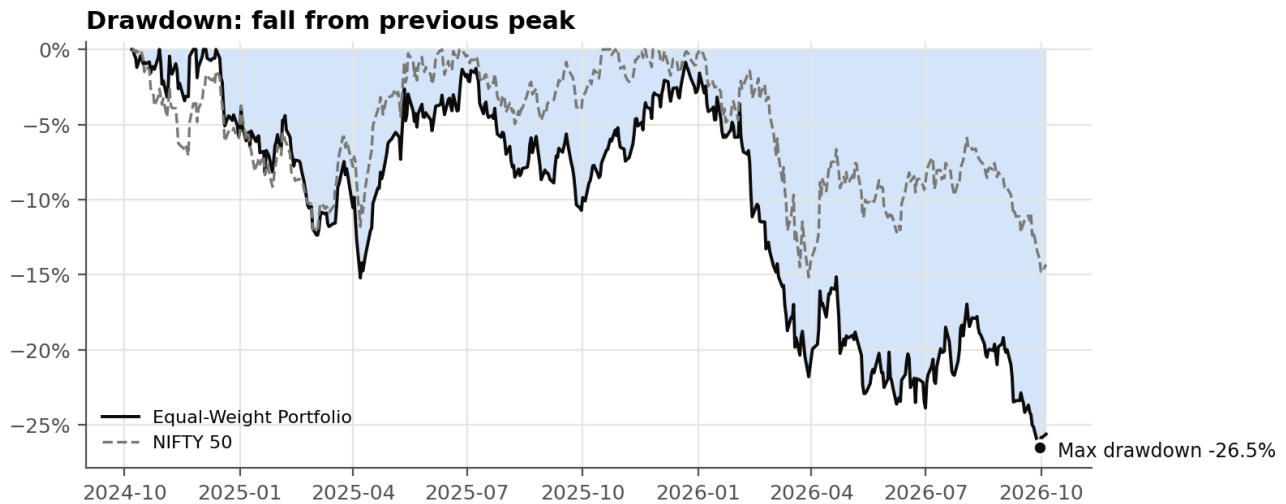


Figure 2. Drawdown of the portfolio. The deepest fall ran from 13 Dec 2024 to 30 Sep 2026.

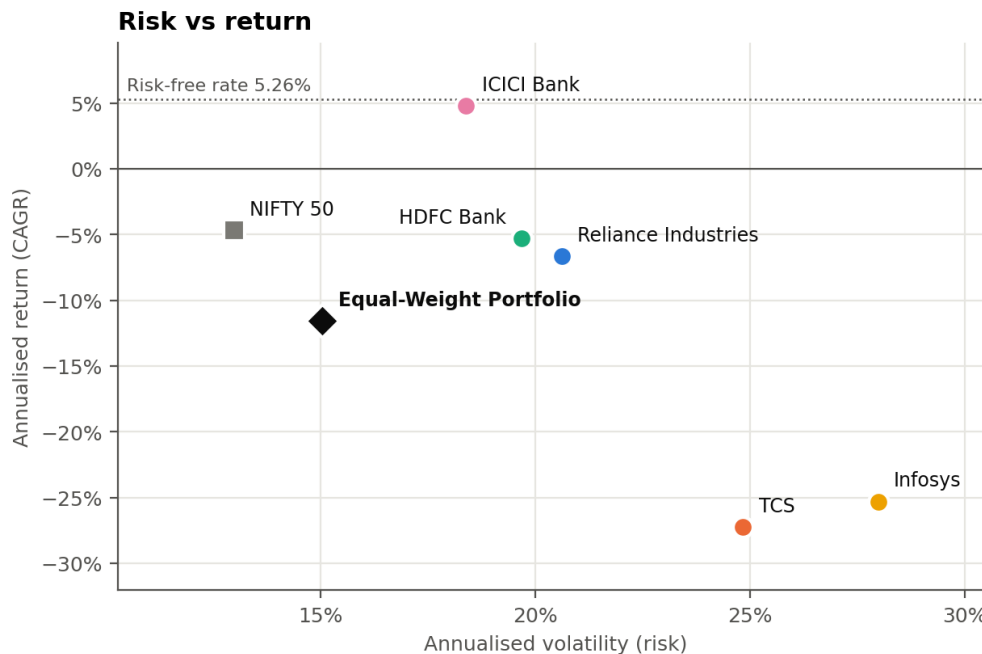


Figure 3. Each security's annualised return against its annualised volatility. Points higher and further left offer more return per unit of risk.

Sharpe Ratio sensitivity to the risk-free rate

The risk-free rate is an assumption, so the Sharpe Ratio is shown at several rates. Rankings that hold across the table do not depend on the exact rate chosen.

Security	rf = 0%	rf = 5%	rf = 6%	rf = 7%
Equal-Weight Portfolio	-0.74	-1.07	-1.13	-1.19
NIFTY 50	-0.30	-0.68	-0.75	-0.83

7. Correlation and diversification

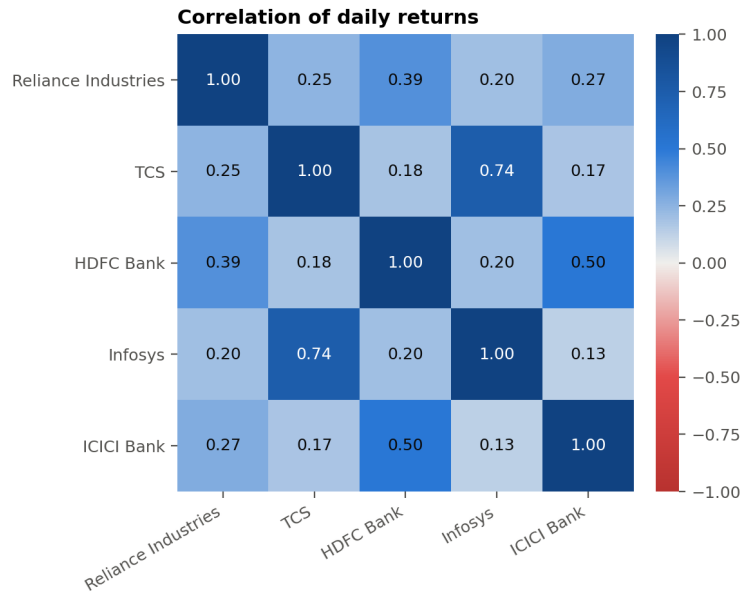


Figure 4. Correlation of daily returns. 1.00 means two stocks always move together; 0 means no linear relationship.

Measure	Value
Weighted average stock volatility (no diversification)	22.3%
Actual portfolio volatility	15.0%
Volatility removed by diversification	32.6%
Diversification ratio	1.48
Average pairwise correlation	0.30
Most correlated pair	TCS / Infosys (0.74)
Least correlated pair	Infosys / ICICI Bank (0.13)

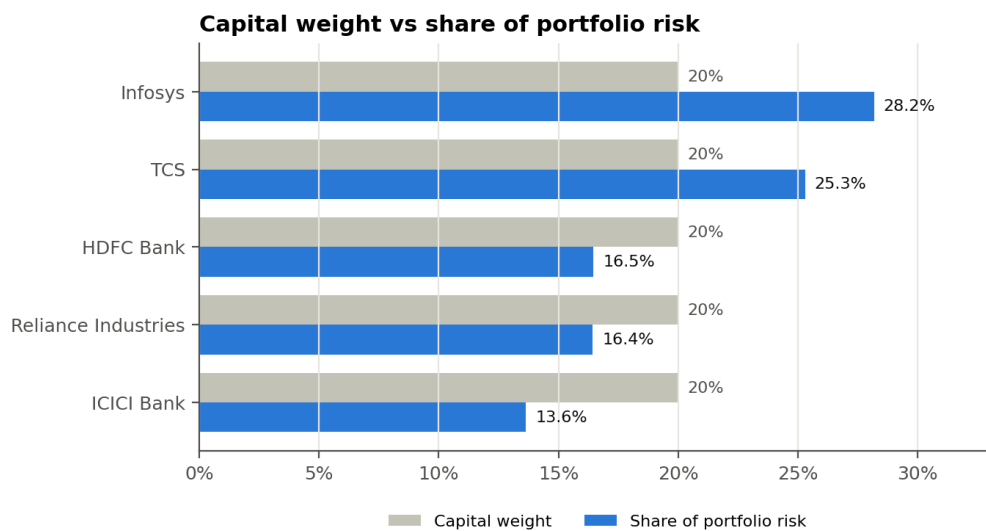


Figure 5. Each stock's 20% capital weight next to its share of portfolio volatility (covariance-based risk contribution).

Stock	Weight	Share of risk	Risk share minus weight	Return contribution (pp a year)
Reliance Industries	20%	16.4%	-3.6 pp	-0.9
TCS	20%	25.3%	+5.3 pp	-5.7
HDFC Bank	20%	16.5%	-3.5 pp	-0.7
Infosys	20%	28.2%	+8.2 pp	-5.0
ICICI Bank	20%	13.6%	-6.4 pp	+1.3

Table 2. Risk contribution adds up to 100% of portfolio volatility. Return contribution is weight x average annualised return, in percentage points; it adds up to the portfolio's average annual return (an arithmetic average, which is higher than CAGR when returns vary).

8. Key findings

- Performance.** Between 7 Oct 2024 and 5 Oct 2026 (500 trading days) the equal-weight portfolio lost 21.6% in total, an annualised return (CAGR) of -11.6%. Over the same days the NIFTY 50 returned -9.0%, so the portfolio underperformed the index by 12.6 percentage points.
- Risk-adjusted return.** Annualised volatility was 15.0% and the Sharpe Ratio was -1.08 using a 5.26% risk-free rate. A negative Sharpe Ratio means the portfolio earned less than a risk-free Treasury Bill over this period. The NIFTY 50's Sharpe Ratio was -0.70. When both ratios are negative, ranking them is not meaningful, because a negative ratio looks less bad simply when volatility is higher; compare total return and drawdown instead.
- Stock comparison.** ICICI Bank had the highest annualised return (4.8%) and TCS the lowest (-27.2%). Infosys was the most volatile stock (28.0%) and ICICI Bank the least (18.4%). ICICI Bank also had the best Sharpe Ratio (0.07).
- Diversification benefit.** If the five stocks had moved in perfect lockstep, portfolio volatility would have been 22.3% (the weighted average of the stocks). The actual figure was 15.0%, so diversification removed 32.6% of risk (diversification ratio 1.48). The average correlation between pairs of stocks was 0.30.
- Concentration.** The most correlated pair was TCS and Infosys (0.74); the least correlated was Infosys and ICICI Bank (0.13). TCS and Infosys are both Information Technology stocks. Sector weights are Banking 40%, Information Technology 40%, Energy & Conglomerate 20%, so five holdings span only 3 sectors. Stocks from the same sector react to the same news, which limits how much risk an equal-weight split can diversify away.
- Risk contribution.** Every stock holds a 20% weight, but risk is not split equally. Infosys accounted for 28.2% of portfolio volatility, the largest share relative to its weight, while ICICI Bank accounted for 13.6%.
- Drawdown.** The worst peak-to-trough fall was 26.5%, from 13 Dec 2024 to 30 Sep 2026. The portfolio had not regained that previous high by the end of the period. The NIFTY 50's maximum drawdown was 15.2%.
- Rebalancing.** The main results assume the portfolio is reset to 20% per stock every day. A buy-and-hold investor who never rebalanced would have had a total return of -20.8%, 0.8 percentage points higher than the rebalanced portfolio. Without rebalancing, weights drift towards whichever stocks have risen, so the two approaches carry different risk over time.

9. Business interpretation

Against the NIFTY 50, the portfolio trailed the index on return while taking more risk; in hindsight a NIFTY 50 index fund would have served a client better over this period. Its beta of 0.95 means that, on average, a 1% move in the index came with a 0.95% move in the portfolio.

Equal capital is not equal risk. Infosys took up 28.2% of portfolio volatility on a 20% allocation. A risk report built only on capital weights would miss this. A natural next step for a portfolio manager would be to compare the current weights with weights that give each stock an equal share of risk.

Diversification worked, but only partly: volatility fell from 22.3% to 15.0%. Because the holdings come from only 3 sectors, adding stocks from sectors with lower correlation to these five (for example consumer staples or healthcare) is the obvious test to run before adding more of the same.

The maximum drawdown of 26.5% is the number a client feels. Volatility describes the average size of daily swings, while drawdown answers the practical question of how much an investor could have been down from the high point,

which is what drives whether clients stay invested.

10. Limitations

- Past returns, volatility and correlations describe one specific period and do not predict future results. Correlations in particular tend to rise during market stress, when diversification is needed most.
- Two years is a short window; it covers only the market conditions that happened to occur in it, so metrics can change materially with a different start or end date.
- Five large-cap stocks from three sectors are not a diversified portfolio in the institutional sense; results are not representative of the wider market.
- Transaction costs, taxes, and the cost of rebalancing are ignored.
- Volatility and the Sharpe Ratio treat upside and downside moves alike and assume returns are roughly normally distributed; real returns have fatter tails.
- A single recent T-Bill yield is used as the risk-free rate for the whole period; the sensitivity table shows how the Sharpe Ratio moves if a different rate is used.
- Adjusted prices from Yahoo Finance are a secondary source and may contain occasional errors; the raw download is saved so it can be checked against exchange data.

11. Conclusion

Over 7 Oct 2024 to 5 Oct 2026 the equal-weight portfolio of five NSE large caps lost 21.6% (-11.6% a year) with 15.0% annualised volatility, a Sharpe Ratio of -1.08 and a worst drawdown of 26.5%. The NIFTY 50 returned -9.0% with a Sharpe Ratio of -0.70. Holding five stocks cut volatility by 32.6% compared with holding perfectly correlated assets, but sector overlap and unequal risk contributions show that an equal-weight split is a starting point rather than a balanced portfolio. These results describe one historical period and are not a forecast.

12. Tools and technologies

Python 3; pandas for data handling and return calculations; NumPy for the covariance and risk attribution maths; Matplotlib for charts; yfinance for data download; ReportLab to generate this report. The full pipeline runs with one command and saves every intermediate table as CSV, so each number here can be checked.

Sources

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